

ERD ASSIGNMENT

WitleShop Online Retail System

WitleShop (Pty) Ltd is a South African online retail company selling electronics, clothing and home appliances through web and mobile platforms. The database must manage customers, products, orders, payments, deliveries and suppliers

Identified Entities

Entity	Purpose
Customer	Stores registered customer information and customer details.
Address	Stores the delivery addresses registered by customers.
Product	Stores products sold by WitleShop.
Category	Groups products into categories.
Supplier	Stores suppliers that provide products.
Order	Stores customer orders.
Order_Item	Associative entity connecting orders and products.
Payment	Stores the payment record for an order.
Delivery	Stores delivery information for an order.

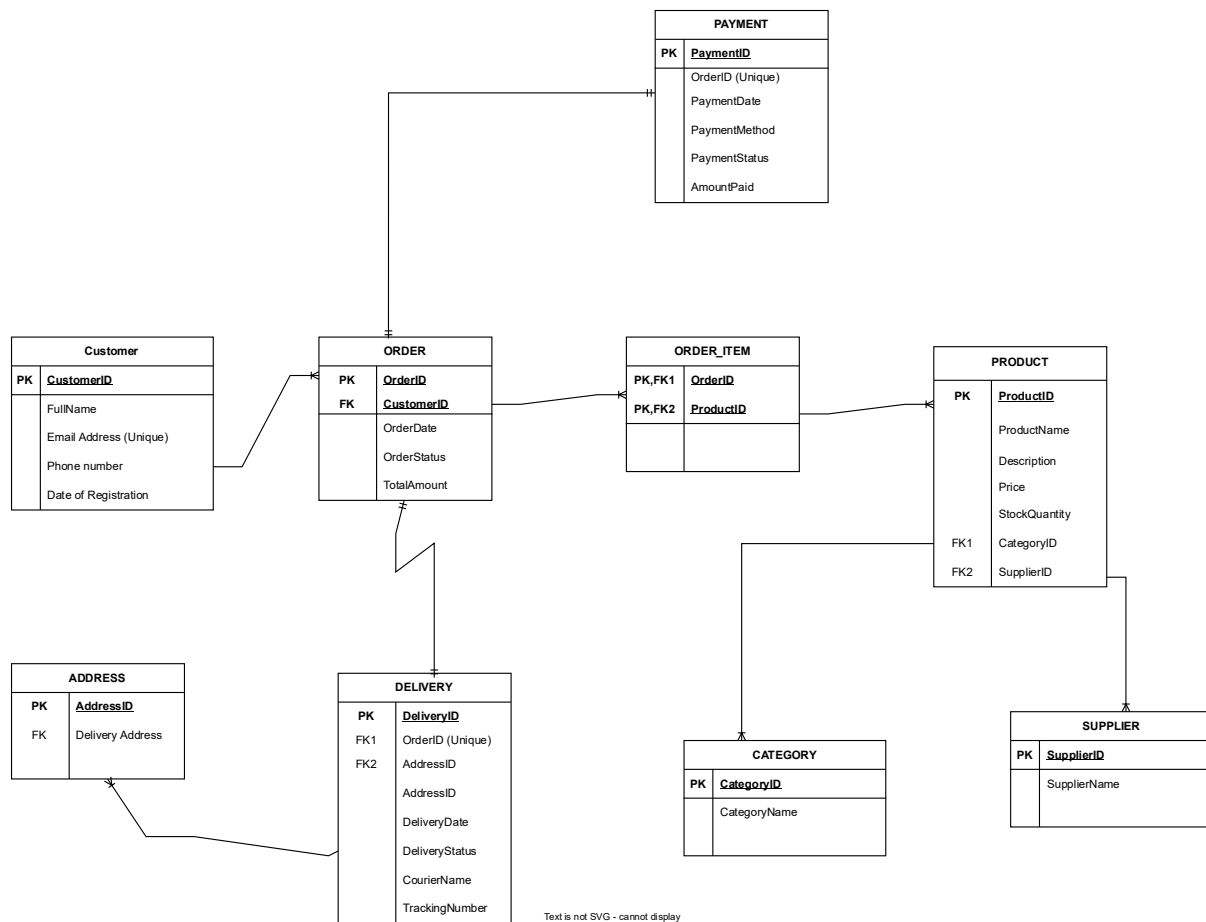
Entities, Attributes, Primary Keys and Foreign Keys

Entity	Attributes	Keys
CUSTOMER	FullName PhoneNumber DateOfRegistration	CustomerID (PK) Email (Unique)
ADDRESS	DeliveryAddress	AddressID (PK) CustomerID (FK)
CATEGORY	CategoryName	CategoryID (PK)
SUPPLIER	SupplierName	SupplierID (PK)
PRODUCT	ProductName Description Price StockQuantity	ProductID (PK) CategoryID (FK) SupplierID (FK)
ORDER	OrderDate	OrderID (PK)
	OrderStatus TotalAmount	CustomerID (FK)
ORDER_ITEM	—	OrderID (PK, FK) ProductID (PK, FK)

PAYMENT	PaymentDate PaymentMethod PaymentStatus AmountPaid	PaymentID (PK) OrderID (FK, Unique)
DELIVERY	DeliveryDate DeliveryStatus CourierName TrackingNumber	DeliveryID (PK) OrderID (FK, Unique) AddressID (FK)

Relationships and Cardinality

Relationship	Cardinality	Explanation
Customer — Address	1:M	One customer can have multiple delivery addresses. Each address belongs to one customer.
Customer — Order	1:M	One customer can place many orders. Each order belongs to one customer.
Category — Product	1:M	One category can contain many products. Each product belongs to one category.
Supplier — Product	1:M	One supplier can supply many products. Each product is supplied by one supplier.
Order — Product	M:N	An order can contain multiple products, and a product can appear in multiple orders. ORDER_ITEM resolves this M: N relationship.
Order — Order_Item	1:M	One order can have many order-item records.
Product — Order_Item	1:M	One product can occur in many order-item records.
Order — Payment	1:1	Each order must have one payment record, and each payment belongs to one order.
Order — Delivery	1:1	Each order must have one delivery record, and each delivery belongs to one order.
Address — Delivery	1:M	A delivery must use one of the customer's registered addresses; an address can be used for multiple deliveries.



7. Business Rules Implemented

- Customers must register before placing an order.
- Customer email addresses are unique.
- A customer may have multiple registered delivery addresses.
- A customer may place multiple orders, but each order belongs to one customer.
- Each product belongs to one category, while a category can contain many products.
- Each product is supplied by one supplier, while a supplier can supply many products.
- An order may contain multiple products, and a product may occur in multiple orders.
- Each order has one payment record, and each payment belongs to one order.
- Each order has one delivery record, and each delivery belongs to one order.
- A delivery is linked to a registered customer delivery address

Conclusion

The completed ERD provides a structured database design for WitleShop's online retail system. It identifies the required entities and their attributes, establishes primary and foreign keys, and represents the required 1:1, 1:M and M: N relationships. The design also uses ORDER_ITEM to properly resolve the many-to-many relationship between orders and products